

CSTN, or the Color Super Twisted Nematic Display, originally developed by Sharp Electronics. The CSTN used colour filters for red, green and blue colours to be displayed. Even though in its earlier years CSTN suffered from slower response times and ghosting, recent advancements in the technology allow a 100 ms response time, 140 degree viewing angle and high quality colour reproduction rivalling that of a TFT display – all at about half the cost.

A few other types of STN displays are:

- ◆ FRSTN – Fast Response STN
- ◆ FSTN – Film Compensated STN. Put a film layer between the display and the second polarizer for increased sharpness and contrast.
- ◆ FFSTN – Double Film STN
- ◆ MSTN – Monochrome STN
- ◆ CCSTN – Color Coded STN

What is a TN TFT

As we know by now, the orientation of the crystals in the TN display is controlled by electric charge. TFT, or Thin-Film Transistor technology puts transistors to use to control the row-column addressing of pixels with much greater precision. Each row and column has a designated transistor and each pixel is a capacitor with insulating liquid crystal between two layers of transparent conductive Indium Tin Oxide layers. They are controlled via transistor switches, one for each pixel. Like the previous technology, this too used a multiplexed addressing system.

Pros

- ◆ **Low Production cost** – The basic form of TN displays do not require complex components like transistors or advanced filters, this is why the production cost is so low, though this does affect its abilities compared to other technologies.



A TN TFT under a microscope. The transistors can be seen at the edges.